

Democratizing a Large Over Ground Virtual Suite (Log_{vs}) Involving Human Interaction for Rehabilitation Sciences and Practice

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ABSTRACT

Virtual environments (VE) used within rehabilitation-based virtual reality (VR) platforms often limit their social environmental context and the ability to move naturally within the VE. This technical note presents an evolving VR platform offering virtual human pedestrian (VP) interaction and highlights specific developments underway and future directions for use in research and practice. The platform called the Large Over Ground Virtual Suite or Log_{vs} uses affordable, mobile VR technology involving a VE with multimodal navigation options (biped and wheeled locomotion) while manipulating VP characteristics (e.g., emotions, expressions, physical interactions). The overall goal of the evolving platform is to democratically enhance interdisciplinary study, assessment, and training of different aspects of social interaction (social cognition, mobility, multiple sensory-motor interactions) to cater to rehabilitation research and clinical needs more effectively. In addition to detailing the platform and examples of current uses and developments, this work serves to also comment on the need for greater sharing of immersive technology resources across the rehabilitation sciences community.

Keywords: Mobility, Gait, Executive Functioning, Social Cognition, Rehabilitation

Index Terms: Applied Computing: Life and medical sciences

1 INTRODUCTION

Combined cognitive, mental, social and physical abilities are crucial for maneuvering through the complex environments of daily life. However, the challenges associated with creating rich, ecological environments while ensuring user safety and standardization of measurements limits the controlled study, assessment, and intervention of one's ability to navigate such complex daily environments. Virtual reality (VR) offers a means to overcome these limitations as well as to greatly motivate and engage the user [1]. In addition, VR can be personalized to the user.

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Yet, using VR in rehabilitation is still perceived to have several barriers related to cost, time and space [2]. Also, rehabilitation professionals are more motivated to use VR tools when they respond to the clients' needs and provide customization and control over situations [3]. VR for rehabilitation still poorly represents combined cognitive, mental, and physical demands of the real world. In addition, locomotion is often represented by means other than over ground walking, such as using joysticks, walking in place or treadmills [4], limiting natural dynamic equilibrium and locomotor adaptations necessary to respond to the social environmental context. One exception is a platform called "Wordplay VR" that uses gamification for over ground mobility skills related to turning and stepping over obstacles while simultaneously solving word puzzles [5]. This platform presently involves a walking area of 3m by 3m. The ability to walk over ground within larger areas with and without walking mobility aids will further open the possibilities for VR research and clinical applications.

In addition, the social context of any VE representing the community must go beyond simply the physical built environment. Social interactions with other pedestrians are crucial to improve the social ecology as well as provide more realistic and complex navigational aspects. Research has advanced how we circumvent pedestrians (e.g., [6]–[8]) and navigate crowds (e.g., [9], [10]), but these need to be integrated in a more generic manner into a platform for rehabilitation sciences and clinical practice. Affective aspects of virtual humans through emotional and communicative expressions will also further enhance the social context of any VR-based tool.

Finally, accurate position and orientation data coupled with physiological measurements are also important for a more general research and clinical VR platform. Such measurements are crucial to understand one's behaviour as they navigate the daily social environment as well as to ensure their safety while moving within the VE.

The main aim of the present technical note is to provide detail (greatly extending on a recent preliminary presentation, [11]) of the on-going development of a VR-based platform by a Team of researchers at the Cirris (Quebec City, Canada) in collaboration with researchers of other research centres (CRIR/Laval, Canada; INRIA/Rennes, France). Specifically, this paper and presentation highlight the promise of a generic, open access, VR-based platform called the Large Over Ground Virtual Suite (Log_{vs}) that integrates virtual humans as pedestrians in an attempt to meet the needs noted in the introduction above with respect to cost, time, space and social context. In addition, some of the limits involved in such development will be discussed. This paper also underscores the democratic nature of the Log_{vs} platform and provides a general comment on the need for greater open access and sharing within the research community and for knowledge exchange.

2 PREVIOUS AND CURRENT DEVELOPMENT OF THE LOG_{vs} PLATFORM

The platform was born out of the need to navigate larger spaces which necessitated the integration of standalone, wireless head mounted displays (HMD). The standalone HMD Oculus Quest 2 (now Meta Quest 2, Meta Platforms Inc., USA), was initially chosen as an affordable, easy-to-use option. Currently both the Meta Quest 2 and Meta Quest Pro HMDs are used, but the platform can load onto other HMDs. The Log_{vs} VE currently involves a public park scenario (see Figure 1) covering over 100 m² with a surrounding street environment (not shown) programmed using the Unreal Game Engine (Epic Games, USA). The full scene of the platform can be used in any large space (e.g., gymnasium; drill hall of a military base) but can also be partitioned to focus on parts of the VE to fit smaller spaces such as a hospital corridor or examination area.



Figure 1: View of part of the VE park interior as seen by the user at the entrance.

Virtual humans presented as other virtual pedestrians (VPs; older/younger, male/female) have been programmed using MetaHuman with either stock standing postures or locomotor animations from previous data collections (e.g., [12]). Locomoting VPs can walk towards the user at a set velocity along a straight line, change direction at a controlled angle and position within the VE or physically interact with the user by reacting to their movements and blocking their path for a predetermined period before allowing collision avoidance. Such physical and social contexts evoke complex trajectory planning found to increasingly solicit executive functioning [13].

Currently, VPs can also express facial emotions that are triggered by the users position with respect to the VP. Action unites manipulated in the Unreal game engine to morph into different intensities of a wide variety of facial emotions have been arranged in collaboration with P. Jackson of our Log_{vs} team according to his work [14]. Examples of such facial emotion presentations are shown in Figure 2.



Figure 2: Examples of two of the emotions that can be expressed by the virtual pedestrians. Left: sad with mouth closed; Right: surprise with jaw drop.

The current version of the Log_{vs} platform only involves single pedestrian interaction. However, collaborations are underway with a group of researchers at INRIA, Rennes (A-H Oliver; J. Pettr ) to integrate crowd dynamics (intelligent interactions between virtual humans and the user) into Log_{vs}. This will greatly enhance the social context of the platform and provide even more realistic means to study, assess and train navigational behaviour and the related cognitive-motor abilities required for different populations across the lifespan.

In addition, other locomotor modes involving wheeled mobility are being currently investigated in order to allow users to navigate the VE in a manual wheelchair or on a tricycle (the latter for a basic research project currently being collected). In close collaboration with F. Routhier of the Log_{vs} team, wheelchair locomotion will open the doors to both fundamental research on complex navigation as well as to developing better means of training wheelchair skills to new users. Figure 3 below shows the user's first-person view of their wheelchair and their avatar where wheelchair displacement and related wheel revolutions are synchronized to the displacement of the HMD while avatar arm movement is synchronized to HMD hand tracking. Development is still in early stages.

Finally, a user Windows- or Android-based interface allows the control of different scenario contexts noted above as well as to collect data from the HMD's inertial measurement units (orientation angles) and inside-out tracking (linear position data). This is all carried out using on-line communication with the HMD through a wi-fi link. Data of the orientation and linear position of the virtual pedestrian can also be exported for further analysis.



Figure 3: First-person view of one's avatar in the wheelchair.

It should be noted that data from the IMU and inside-out technology of HMDs like the Quest2 and Quest Pro must be considered carefully. A recent publication from our team [15] has shown that for the Oculus Quest 2 (now called Meta Quest 2) standalone HMD, the criterion validity during static conditions is relatively good with mean errors around 1 cm and 1° along all axes. However, it is a different story for dynamic conditions such as walking where mean errors increased to around 4 cm and 4° with maximum errors as much as 19 cm and 14°. Therefore, while perfectly fine for scene presentation and the user's immersive experience, translation motion capture data as outcome variables may be best provided with additional motion trackers.

3 FUTURE DIRECTIONS

Despite some attempts to create over ground VR-based platforms for cognitive-motor rehabilitation [5], there is a great need to advance in this direction and involve more socially based contexts. Work on crowd dynamics in over ground situations [e.g., 9] is inspiring but needs to be put in the context of rehabilitation. As noted above, the development of the Log_{vs} platform involves an interdisciplinary team of researchers in rehabilitation sciences at the Cirris in collaboration with researchers from other centres. These researchers are pushing the boundaries of the human body functions to be explored through the Log_{vs} platform. In addition to the early developments briefly presented in section 2 above, bodily expressions as well as more realistic facial expressions (e.g., MetaHumanAnimator, Unreal Engine) from the VPs are planned to continue to push interactions beyond just the physical and further enhance the social context of the VE. In this vein, integrating speech will be pursued to enhance communicative interactions (both during standing and while in dynamic states) with VPs. This, along with integrating 3D sound (i.e., beyond the simple presentation through HMD speakers) will allow researchers and clinicians in speech therapy and audiology to use the platform in more dynamic realistic environments. Object manipulation with the upper limbs will also add to the ways the user can interact with the VPs and the VE in general, including the integration of gamification for further motivation. The integration of other physiological measures such as galvanic skin measures, heart rate, respiration can be easily added in the future for parallel behavioural data collection. As also noted, motion capture from the HMD alone is limited and needs to be further explored. We have not yet pushed the limits of using controllers that come with the Quest HMDs that are presently targeted. The continuous consideration of the evolution of related technology (e.g., new HMDs, markerless motion capture systems, etc.) and game engines will obviously be crucial as the platform continues to develop.

We feel that both the use of accessible, relatively low-cost technology as well as an open access approach makes Log_{vs} highly democratic. This is important as resource sharing in VR research in general, but in rehabilitation sciences specifically, is still sparse and programming resources in particular remain an important bottle neck in advancing such research and collaborating. This is why the Log_{vs} platform has been integrated into a GitLab platform and we are currently arranging an open access licensure. Once completed, the goal is to allow open use of executable forms of Log_{vs} by other researchers and clinicians as well as to encourage parallel programming for its continued development with the open agreement that all developments remain accessible to everyone. The logistics of this are not simple, but the repercussions are important as an example of how we can break down institutional and disciplinary boundaries to greatly advance VR-based rehabilitation science together. This open-source approach is only related to the use of the platform and any data collected with the platform must

meet ethical standards. Presently, data are only transferred locally, and data sharing is outside the intended function of the platform.

Finally, clinical implementation of any VR-based platform, Log_{vs} is no exception, remains a significant challenge. Ensuring the acceptability of such a technology, from the early stages of development, is crucial to facilitate its rapid adoption. Focus groups involving 6 clinicians working with neurological clients (occupational therapists, physical therapists, neuropsychologists) highlighted the fact that, although they target components of locomotor navigation in their assessments and interventions, they rarely evaluate or train them directly due to their complexity. Yet, clinicians found that the early versions of Log_{vs} held significant potential as a tool for cognitive and sensorimotor rehabilitation. Log_{vs} may go a long way to help clinicians access such navigational complexity in an affordable, easy-to-use manner. However, continued documentation of the needs and perceptions of clinicians is crucial.

4 SUMMARY OF PRIORITIZED ACTIONS THAT ARE PLANNED

Based on the future directions noted above, we provide below a list of actions planned for Log_{vs}, that can also serve to guide future work on similar platforms.

- Further build multidisciplinary, multi-centre teams.
- Strengthen collaborations with clinicians and other stakeholders.
- Improve expressive and communicative interactions by virtual humans.
- Enhance other sensory experiences (e.g., audio, haptic) for task goals and ambient distractions/perturbations.
- Expand on the tasks that can be carried out, including upper limb functions and multi-tasking.
- Integrate other measures of motion and physiology.
- Continue to advocate and work towards open licensure and shared resources and to constantly assure ethical use of data.

5 CONCLUSION

In conclusion, the Log_{vs} VR platform offers a novel means for studying, assessing, and training different functions (cognitive, mental, physical) within the context of community navigation involving virtual human pedestrians. This approach enriches the social context of the VE while offering both static and dynamic (i.e., with over ground displacements) interactions. The targeted infrastructure is purposely affordable and mobile, and a user-friendly interface attempts to simplify the varied uses of the platform. The Log_{vs} platform is set-up to encourage as much as possible co-development and continued evolution within an open access environment to help democratize VR-based research and practice for the rehabilitation sciences.

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REFERENCES

- [1] B. Mouatt, A. E. Smith, M. L. Mellow, G. Parfitt, R. T. Smith, et T. R. Stanton. The Use of Virtual Reality to Influence Motivation, Affect, Enjoyment, and Engagement During Exercise: A Scoping Review. *Front. Virtual Real.*, 1:2020,
- [2] D. Levac, S. Glegg, H. Colquhoun, P. Miller, F. Noubary. Virtual Reality and Active Videogame-Based Practice, Learning Needs, and Preferences: A Cross-Canada Survey of Physical Therapists and Occupational Therapists. *Games Health J.*, 6(4):217-228, 2017.
- [3] R. Proffitt, S. Glegg, D. Levac, B. Lange. End-user involvement in rehabilitation virtual reality implementation research. *J. Enabling Technol.*, 13(2):92-100, 2019.
- [4] C. Mousas, D. Kao, A. Koiliias, B. Rekabdar. Evaluating virtual reality locomotion interfaces on collision avoidance task with a virtual character. *The Visual Computer*, 37(6):2823–2839, 2021.
- [5] J.M. Finley, M. Gostsis, V. Lympouridis, S. Jain, A. Kim, B. Fisher. Design and Development of a Virtual Reality-Based Mobility Training Game for People With Parkinson's Disease. *Fron. Neurology*, 15(11):577713, 2021.
- [6] M. Gérin-Lajoie, C.L. Richards, B.J. McFadyen. The Negotiation of Stationary and Moving Obstructions during Walking: Anticipatory Locomotor Adaptations and Preservation of Personal Space. *Motor Control*, 9(3):242-269, 2005.
- [7] F. Fiset, A. Lamontagne, B.J. McFadyen. Effects of age on the visuo-locomotor control used to circumvent a virtual pedestrian with different limb movements. *Hum. Mov. Sci.*, 87:103026, 2023.
- [8] A.-H. Olivier, A. Marin, A. Crétual, J. Pettré. Minimal predicted distance: A common metric for collision avoidance during pairwise interactions between walkers. *Gait Posture*, 36(3):399-404, 2012.
- [9] L.A. Meerhoff, J. Pettré, S.D. Lynch, A. Crétual, A.-H. Olivier. Collision avoidance with multiple walkers: Sequential or simultaneous interactions? *Front. Psychol.*, 9, 2018.
- [10] A. Koiliias, M.G. Nelson, C.-N. Anagnostopoulos, C. Mousas. Immersive walking in a virtual crowd: The effects of the density, speed, and direction of a virtual crowd on human movement behavior. *Comput. Animat. Virtual Worlds*, 31(6):e1928, 2020.
- [11] J. Boulo, A.K. Blanchette, A. Cyr, J. Caron-Roberge, P. Jackson, CS Batcho, A. Lamontagne, B.J. McFadyen. LogVS: Developing an affordable and user-friendly virtual reality platform to study, assess and train different aspects of locomotor navigation», presented at WCISVR - 15th International Conference on Virtual Rehabilitation, Montréal, Canada, June 2023.
- [12] A. Deblock-Bellamy, A. Lamontagne, B.J. McFadyen, M.-C. Ouellet, A.K. Blanchette. Virtual reality-based assessment of cognitive-locomotor interference in healthy young adults. *J. NeuroEngineering Rehabil.*, 18(1):53, 2021.
- [13] N. Robitaille, P.L. Jackson, L.J. Hébert, C. Mercier, L.J. Bouyer, S. Fecteau, C.L. Richards, B.J. McFadyen. A Virtual Reality avatar interaction (VRai) platform to assess residual executive dysfunction in active military personnel with previous mild traumatic brain injury: proof of concept, *Disabil. Rehabil. Assist. Technol.*, 12(7):758-764, 2017.
- [14] P.L. Jackson, P.-E. Michon, E. Geslin, M. Carignan, D. Beaudoin. EEVEE: the Empathy-Enhancing Virtual Evolving Environment. *Front. Hum. Neurosci.*, 9, 2015.
- [15] J. Boulo, A.K. Blanchette, A. Cyr, B. J. McFadyen. Validity and reliability of the tracking measures extracted from the Oculus Quest 2 during locomotion, *Comput. Methods Biomech. Biomed. Eng. Imaging Vis.* Published Online: 18 Nov 2023.